

PubMed AI Summary

Topic: paracetamol

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Paper 1

****Title****

Differentiating Hepatic and Renal Toxicity Reveals CYP-Independent Mechanisms of Acetaminophen-Induced Acute Kidney Injury

****Evidence Level****

Animal/in-vitro study (genetically modified mouse model)

****Study Size & Population****

Male and female CYP2E1^{flox/flox} mice treated with AAV8-TBG-Cre for liver-specific CYP2E1 deletion; sample sizes per group not specified in abstract. Mice received a single acetaminophen overdose (600 mg/kg) and were evaluated up to 48 hours.

****Key Findings****

- Liver-specific CYP2E1 deletion successfully reduced hepatic NAPQI formation and hepatotoxicity, confirming the model's validity.
- Despite hepatic protection, both male and female mice developed progressive acute kidney injury (AKI) evidenced by marked elevations in BUN and creatinine, tubular vacuolation, and strong induction of KIM-1 and osteopontin at 48 hours.
- Female mice lacking renal CYP2E1 with no detectable renal protein adducts still developed AKI, demonstrating that kidney injury occurs via CYP-independent mechanisms (potentially involving p-aminophenol/PAP) during the later phase of toxicity.

****Adverse Events / Safety Signals****

Acetaminophen overdose (600 mg/kg) caused severe hepatotoxicity (in controls) and progressive AKI (BUN/creatinine elevation, tubular vacuolation, cell death) independent of hepatic or renal CYP2E1 activity.

****Study Limitations****

Preclinical mouse model; results may not directly translate to human acetaminophen pharmacokinetics, dosing, or comorbidities. Sample sizes and statistical power not reported in abstract. Single time-point (48h) focus on delayed injury; earlier mechanisms not fully characterized. Funding source/conflicts not

stated in abstract.

****Relevance to "paracetamol"*****

This study suggests that paracetamol-induced kidney injury in overdose is a kidney-intrinsic process independent of CYP-mediated bioactivation, implying that renal protective strategies must target non-CYP pathways (e.g., p-aminophenol) rather than solely focusing on hepatic NAPQI reduction.

Paper 2

****Title****

FDA Approves First Acetaminophen and Naproxen Sodium Combination Drug

****Evidence Level****

Not applicable (news article/regulatory announcement; no original research data presented)

****Study Size & Population****

Not applicable (no study population described)

****Key Findings****

- The FDA has approved the first fixed-dose combination product containing acetaminophen and naproxen sodium.
- Specific brand name, dosage strengths, indication, and pivotal trial data are not provided in the abstract text.

****Adverse Events / Safety Signals****

None reported in abstract.

****Study Limitations****

Not applicable (news summary); abstract lacks clinical trial data, efficacy/safety results, dosing details, or prescribing information.

****Relevance to "paracetamol"*****

Pharmacy professionals should anticipate a new fixed-dose combination analgesic (acetaminophen + naproxen sodium) entering the market, requiring counseling on dual-mechanism dosing, maximum daily acetaminophen limits, and NSAID-specific risks (GI, CV, renal).